

Anupam Agrawal
Spring 2024 - Fall 2024
Faculty Performance Evaluation Review Form
Associate Professor
anupam@tamu.edu

Degrees

2008	Ph.D., Operations Management, INSEAD France
2006	M.S., Technology Management, INSEAD, Fontainebleau, France
1994	M.B.A., Business Administration, Indian Institute of Management
1992	BEng, Aerospace Engineering, Indian Institute of Technology, Kanpur, India

Professional Licensures

Input Form									
Effective Date	Expiration Date	Title	Organization	License Number (if applicable)	Month/Year Recertified	State of Issue	Country (if international licensure)	Type (if applicable)	Description
2017-01-01		Social and Behavioral Research Investigators	TAMU IRB						IRB Certification for working with human subjects

Teaching (Custom Section)

Teaching Input Section											
Course Title	Course	Credit Hours	Lab Hours	Instruction Mode	Lecture Hours	Student Credit Hours	Enrollment	If co-taught, % contributed to course	Descriptive (if applicable)	Semester	Multidisciplinary Collaboration Activities Involved
SOURCING AND PROCUREMENT	SCMT 335 502	3		TR	3	75	25			Fall 2024	
SOURCING AND PROCUREMENT	SCMT 335 501	3		TR	3	57	19			Fall 2024	
SOURCING AND PROCUREMENT	SCMT 335 503	3		TR	3	72	24			Spring 2024	
SOURCING AND PROCUREMENT	SCMT 335 502	3		TR	3	63	21			Spring 2024	

Student Evaluations Data

Course	Total Surveyed/Enrolled	Total Completed Survey	Q1 Avg (1-4)	Q1 StDev	Q2 Avg (1-3)	Q2 StDev	Q3 Avg (1-4)	Q3 StDev	Q4 Avg (1-4)	Q4 StDev	Q5 Avg (1-4)	Q5 StDev	Q6 Avg (1-6)	Q6 StDev	Q7 Avg (1-6)	Q7 StDev	Q8 Avg (1-5)	Q8 StDev	Q9 Avg (1-3)	Q9 StDev	Q10 Avg (1-3)
SCMT 335 502	25	12	3.9167	0.2764	2.9167	0.2764	3.8333	0.3727	3.8333	0.3727	3.7500	0.4330	4.9167	0.2764	5.6667	0.4714	4.9167	0.2764	3.0000	0.0000	2.8333
SCMT 335 501	19	11	4.0000	0.0000	3.0000	0.0000	3.9091	0.2875	3.6364	0.4810	3.8182	0.5750	4.9091	0.2875	5.6364	0.4810	4.8182	0.3857	3.0000	0.0000	2.9091
SCMT 335 502	21	8	3.7500	0.4330	2.8750	0.3307	3.5000	0.7071	3.6250	0.6960	3.7500	0.6614	4.3750	0.6960	5.2500	0.9682	4.0000	1.5000	2.8750	0.3307	2.7500
SCMT 335 503	24	12	3.8333	0.3727	3.0000	0.0000	3.8333	0.3727	3.7500	0.5951	4.0000	0.0000	4.7273	0.4454	5.5000	0.8660	4.4167	0.8620	2.8333	0.3727	3.0000

Publications, Conference Proceedings, Patents and Creative Products/Innovations

Journal Article

Submitted

Applied or Integration/Application Scholarship

- Agrawal, Anupam, and Ujjal Mukherjee. 2022. "The Impact of Buyer's Organization Structure on Suppliers' Quality," June. [Scholarly Type: Applied or Integration/Application Scholarship]

In Progress

Basic or Discovery Scholarship

- Majumdar, Mayukh, Anupam Agrawal, Arun Sen, and Chelliah Sriskandarajah. 2022. "Cost and Quality Performance of Accountable Care Organizations: An Empirical Investigation," September. [Scholarly Type: Basic or Discovery Scholarship]
- Majumdar, Mayukh, Anupam Agrawal, Arun Sen, and Chelliah Sriskandarajah. 2022. "Cost and Quality Performance of Accountable Care Organizations: An Empirical Investigation," September. [Scholarly Type: Basic or Discovery Scholarship]
- Biswas, Indranil, and Anupam Agrawal. 2022. "Effect of Leadership and Contract Sequence on Three-Echelon Supply Chain Coordination," September. [Contribution Level: Equal] [Scholarly Type: Basic or Discovery Scholarship]
- Majumdar, Mayukh, Anupam Agrawal, Chelliah Sriskandarajah, and Bala Shetty. 2020. "Safety Stock Allocation in an Online Retailing Network: A Stochastic Optimization Approach," July. [Scholarly Type: Basic or Discovery Scholarship]
- Majumdar, Mayukh, Anupam Agrawal, Arun Sen, and Chelliah Sriskandarajah. 2022. "Utilizing Specialists and Nurse Practitioners on Improving Accountable Care Organizations' Efficiency: An Optimization Framework," September. [Contribution Level: Contributing] [Scholarly Type: Basic or Discovery Scholarship]
- Majumdar, Mayukh, Anupam Agrawal, Bala Shetty, and Chelliah Sriskandarajah. 2022. "Safety Stock Allocation in an Online Retailing Network: A Stochastic Optimization Approach." [Contribution Level: Contributing] [Scholarly Type: Basic or Discovery Scholarship]

Applied or Integration/Application Scholarship

- Kim, Youngsoo, Dharma Kwon, and Anupam Agrawal. 2022. "Strategic Investment in Shared Suppliers with Quality Deterioration," January. [Scholarly Type: Applied or Integration/Application Scholarship]
- Mukherjee, Ujjal, and Anupam Agrawal. 2022. "Evolution of Organizational Routines: A Multi-Method Study," June. [Scholarly Type: Applied or Integration/Application Scholarship]
- Mukherjee, Ujjal, and Anupam Agrawal. 2022. "Information Integration in Customized Complex Product Manufacturing: An Empirical Investigation of European Bus Building Industry," May. [Scholarly Type: Applied or Integration/Application Scholarship]
- Agrawal, Anupam, and Paolo Letizia. 2020. "Customization and Product Returns - An Empirical Investigation," January. [Scholarly Type: Applied or Integration/Application Scholarship]
- Youn, Seokjun, Anupam Agrawal, Subodha Kumar, and Chelliah Sriskandarajah. 2022. "Selecting Healthcare Providers for Bundled Payments in Healthcare Services," January. [Scholarly Type: Applied or Integration/Application Scholarship]
- Shukla, Dharendra, Alok Raj, and Anupam Agrawal. 2023. "Cultural Distance in Cross-Border Alliance Networks and Firm Innovation: The Contingent Role of Network Governance." [Author Order Number: 3] [Contribution Level: Equal] [Number of Co-Authors: 3] [Scholarly Type: Applied or Integration/Application Scholarship]

14. Youn, Seokjun, Anupam Agrawal, Subodha Kumar, and Chelliah Sriskandarajah. 2022. "Selecting Healthcare Providers for Bundled Payments in Healthcare Services," January. [Contribution Level: Contributing] [Scholarly Type: Applied or Integration/Application Scholarship]

Professional Interests

Professional Interests	
Professional Interests	

Service

Service								
Committee Name/Activity	Organization	Service Type	Service Role	Regionality	Brief Description/Impact (if applicable)	Start Semester	End Semester	Multidisciplinary Collaboration Activities Involved
Judge for 2024 Best Working Paper for INFORMS TIMES	INFORMS	Professional	Reviewer			Fall 2024	Fall 2024	Yes
University Academic Freedom Council		University	Committee Member			Fall 2024	Ongoing	Yes
University Honors & Undergraduate Research Advisory Committee		University	Committee Member			Fall 2024	Ongoing	Yes
INFORMS MSOM 2024 SIG Evaluation	MSOM	Review Panel	Reviewer			Spring 2024	Spring 2024	Yes
Judge for 2023 Best Working Paper for INFORMS TIMES	INFORMS	Professional	Reviewer			Fall 2023	Spring 2024	Yes
Faculty Grievance Committee		University	Member			Spring 2021	Ongoing	
	Anjali School (for destitute children)		Volunteer			Spring 2020	Ongoing	
	Managment Science, Manufacturing and Service Operations Management, European Journal of Operational Research, Production and Operations Management, Journal of Operations Management	Professional	Reviewer/Referee			Spring 2008	Ongoing	
	Laxmi Ashram School for Girls		Volunteer			Spring 2008	Ongoing	

Professional Affiliations and Memberships

Professional Affiliations and Memberships				
Organization Name	Membership Status	Description	Start Semester	End Semester
INFORMS	Member		Summer 2008	Ongoing
:POMS	Member		Summer 2008	Ongoing

Teaching, Research and/or Other Scholarly or Creative Activities, and Service Impact Statement (Faculty Evaluation Review)

TRS / Creative Activities Statement	Semester
1. My research impacts society in two ways. First, my empirical research with firms such as Tata Motors, Tata Daewoo, BMW, and Knorr Bremse which focused on procurement, sourcing of modular components and learning between buyers and suppliers, has informed the practice of supply chain at these firms and made an impact in improving business practices directly. Second, my research on societal issues, such as the coronavirus pandemic, helps society at large by uncovering possible solutions and applications for addressing the pressing needs of the society. 3. My teaching is making a positive impact on society by addressing the critical issue of trust and morality between buyers and suppliers. Specifically, my own case study "Infosys: The Moral Compass" addresses the issue of how businesses can benefit from being good citizens. 2. My service impacts society directly. I am very actively involved with the education of underprivileged children, specifically with (a) Laxmi Ashram, an organization in Kausani hills in India, that works for the education of underprivileged girls, and (b) Anjali School, a school in the ancient city of Varanasi in India, that focuses on the education of the poorest of the poor. I actively work with Professional socities such as M&SOM and help suggest policies to make society more inclusive, including dealing with sexual harassment and racism.	Fall 2024
Research, Teaching and Service Statement RESEARCH My research is of interest to both academics and practitioners in the field of supply chain management. My research spans the business and engineering domains. My primary research interest is in sourcing and retailing management, specifically in four areas of enquiry: (i) How can firms benefit from better raw materials sourcing? (ii) How do buyers and suppliers in a supply chain learn? How does this learning depreciate? (iii) How can retailers develop better returns policies for their customers? (iv) How do traditional contracting schemas need to be modified in real-life situations? <i>My associated (interdisciplinary) research interest is in the manufacturing and new product development domain, specifically nanomachining and materials.</i> To accomplish this research, I draw on my academic training at INSEAD in empirical and analytical methods, my ten years of professional experience as a supply chain manager, and my engineering training to identify issues relevant to operating managers. To do so, I strive to develop an in-depth understanding of the relevant business and technology contexts, enabling me to access unique operational information to: - Create innovative datasets that can be examined using a variety of empirical methods. - Use game theoretical models and simulation to explore optimal strategies that can inform the empirics and vice versa. In 30 peer reviewed published research papers and several working papers spanning management and manufacturing, I have 2115 citations of my work, with an H-index of 23 and an I-index of 27. Raw Material Management in Supply chains This stream of research is informed by my close association with four automotive firms during my dissertation. The central idea is that with an increased focus on core business activities and the outsourcing of non-core activities, firms have slowly distanced themselves from some of the sources of value in their upstream supply chains, like managing raw material. Building close relationships with upstream suppliers' suppliers, such as raw material suppliers, can be valuable for firms. In three papers (w/ De Meyer and Van Wassenhove, in <i>M&SOM</i> 2014 and <i>CMR</i> 2014; solo authored in <i>EJOR</i> 2014), my co-authors and I explore how a firm's raw material sourcing	Fall 2024

knowledge can be a strategic resource and how firms can acquire and integrate this knowledge. By examining the sourcing experiences of four firms in four different countries (in Germany, France, India, and South Korea) in the automotive industry, I identify the raw material sourcing knowledge-related parameters, and analytically explore the concept of the sourcing hub, a collaborative center involving the firm, its suppliers, and the raw material suppliers as the principal mechanism for acquiring and deploying raw material sourcing knowledge in order to manage value in upstream sourcing. My results show that when a firm brings its suppliers and suppliers' suppliers together, value is created by pooling knowledge: information about demand, process improvements, raw material sourcing, and design complexity reduction is exchanged. This can help build an efficient supply chain.

Learning and Forgetting between Buyers and Suppliers

In this stream of research, I explore how knowledge development can impact operational processes, specifically quality management. My six papers in this area (four empirical and two analytical) are informed by my close association with one of the largest automotive firms in Asia. This post-dissertation work has enabled me to prepare several archival datasets based on operational initiatives undertaken at this firm and also analyze these processes mathematically, which has helped create a rich pipeline for current and future research projects.

In two analytical papers (W/ Kwon, Xu, and Muthulingam, *Mgt Sc. 2016*, and w/ Kwon, Kim and Muthulingam, *POM 2016*), we investigate the impact of the interplay between learning effects and externalities on competitive investments with uncertain returns. We propose optimal strategies for firms to invest in quality improvement at their suppliers when the benefits of such investments have the potential to spill over to other firms who source from the same suppliers. We find that an increased rate of learning can hasten the investment by competitive firms under appropriate conditions. The spillover of quality improvement from one firm to the other tends to reduce the first investment threshold and the time to first investment. In contrast, the spillover of information about the supplier's true quality increases the first investment threshold and can affect the time to first investment.

In an empirical paper (w/ Mukherjee, *under review*), we analyze the supplier's response to the buyer's organizational arrangements by analyzing primary panel data from a very interesting real-life experiment and simulating the contingencies. The deployment of an organic structure in one plant enabled firm engineers to work on long-term goals with suppliers, thus enabling the creation and use of conceptual knowledge. As a result, this plant showed a sustained improvement in the quality of its sourced components, even when the sourcing was from the same suppliers who supplied to the other plant. In two empirical papers (w/ Muthulingam - *M&SOM 2015*, and *M&SOM 2016*), we use data on quality improvement initiatives undertaken at the vendors of an automotive OEM to examine how quality knowledge at vendors can be retained. We find that vendors *forget* quality learning gained from quality improvement initiatives (induced learning) as well as from learning-by-doing. In an extension (w/ Muthulingam and Mukherjee, *POM 2020*), we explore how forgetting of learning and spillover at vendors are related.

This research contributes to existing literature by enhancing our understanding of how quality knowledge is developed and retained within supply chains. This research is also managerially relevant because it shows how one can build and sustain quality improvement at suppliers, even when your suppliers supply to your competition.

Returns in Retail

In this stream of research, I explore the impact of return policies for customers on sales, returns, and profitability for retailers. In the first paper (w/ Ertekin *M&SOM - accepted*) we analyze primary data from a multichannel retailer and find that customers adopt different return behavior across retail channels: customers tend to avoid using the policy close to the end of the return period if purchased online whereas they don't mind this at all if purchased in Brick and mortar (BM) stores, most likely due to BM stores' instant gratification. We also find that if retailers restrict their return time window, customers tend to respond by returning things earlier. We demonstrate such behavior more strongly for online purchases than for BM store purchases. In a second paper (w/ Letizia, *under preparation*) I am working with another primary dataset obtained from a multichannel retailer who is experimenting with its return policies across products in different geographies.

This research contributes to existing literature by enhancing our understanding of behavioral differences in customers across channels and across geographies. This research is also managerially relevant, since it shows firms should carefully evaluate the return behavior of their customers at each channel; otherwise, retailers may end up with a costly return policy change that hurts overall profitability.

Sourcing and Contracting

My research on contracting between buyers and suppliers is informed by my empirical work and analyzes scenarios that have not been explored in the current literature. In the first paper in this arena, (w/ Hasija and Bhattacharya, *POM 2016*), we analyze the knowledge supply chain and explore the acquisition and licensing of a cost-reducing innovation. We show that IP acquisition and licensing critically depends on the degree of the cost-reducing innovation and patent intermediaries serve to make markets more efficient. In the second paper (w/ Rajapaksha and Muthulingam, *POM 2017*), we explore real-life data and also analytically explore how product modularity decisions of a buyer affect its contracting with its suppliers.

This research contributes by enhancing our understanding of how existing theoretical explanations can be enriched to include real life situations. This research is also managerially relevant because it shows how one can incentivize people, and how one can establish long-term contracts with suppliers.

Nanomachining and Manufacturing Innovations

My research in the manufacturing domain is focused on new technologies for product development. I focus on developing simulations and analyzing data collected in laboratories and the field to assess how nanomachining and emerging materials can help firms develop better and more robust products. In my review paper, (w/ Goel, Luo and Reuben, *IJMTM*, 2015), we analyze the extant methods, explore questions that are open, and suggest opportunities for future research in the area of molecular dynamics simulation ([this paper has over 450 citations](#)). In a recent paper (w/ Pana, Zhoua, Yana, Wang, Guo and Goel, PE, 2021) we discuss advances in the realm of machining theory, experimental design and Artificial Intelligence related to ground surface roughness prediction. This research helps to improve our understanding of how firms can address precision machining challenges for new materials in their new product development goals. In another recent paper (w/ Vishwanathan et al., *MSSP*, 2024), we explore how we can predict anomalies in real-time in high temperature environments, a crucial step towards sustainable material and manufacturing digitalization.

TEACHING

To me, teaching is not merely a profession—it is my passion. It was with the goal of becoming a teacher that I made the decision to quit my industry job of 10 years and enroll in the PhD program at INSEAD. The joy of teaching and training young minds spills over into everyday interactions, and I have become a better person and a communicator in my new role as a teacher.

My research motivates my teaching. I am interested in empirical applications—I love to go out into the field, work with firms, and then analyze issues in various areas of supply chain management. I bring this real world research into the classroom, and believe that research-centered teaching helps produce better informed graduates and managers who can more effectively lead firms. Relevant research examples bring excitement and freshness into the classroom, and also feed back into research by bringing new perspectives from students. I integrate key research issues with relevant pedagogical material, and organize the course in distinct modules, where each module contributes to one of the learning objectives. For instance, in my supply chain course, the modules are Uncertainty Management, Supply Chain Design, Risk Pooling, Sourcing, Supply Chain Coordination, and Binding Concepts. While the uncertainty module helps to provide a link to analytical tools such as inventory models, the design module helps students to understand the concepts related to postponement and product design and customization. The sourcing module focuses on buyer-supplier coordination, and I weave in my own research on sourcing in this module. The coordination module focuses on an appreciation of the bullwhip effect and double marginalization, and the binding module is a journey into supply chain issues related to ethical behavior between buyers and suppliers, humanitarian logistics, and product returns. Such a design facilitates the understanding of specific concepts and also helps relate these concepts to the overall course objectives. I utilize a similar course design for my courses on Project Management and Business Process Management.

As a teacher, I hold students to high standards. I have found that challenging students and holding them to a high level of performance brings out the best in them. I expect my students to be able to articulate meaningful observations in their own words, to define key terms accurately, to dig beneath the surface in answering questions, to be very good at problem solving and not shy away from numbers, to read and analyze case studies carefully, to situate business problems of the cases in a broader context, and to grapple with the contradictions and complexities of historical cause and effect. Succeeding at these tasks takes time and effort, and students (some more quickly than others) discover that my courses are not what some would call a “blow off course,” but rather an area where they must work very hard to truly excel.

I also work very hard at teaching. I organize my courses around case studies (many of them written by me) and facilitate class discussion by providing a framework that includes discussion questions with a write-up to be prepared before class. This helps students to prepare in advance for the discussion questions and helps them to develop a deeper understanding of the underlying concepts. In addition to the classroom sessions, I strive to reinforce the concepts taught by actively reaching out to students. After each session, I provide a synopsis of the key concepts and additional material that reinforces the material covered in class. I also grade their write-ups myself. These processes of pre-work, active involvement of students in the class discussion, and post-discussion summary and feedback after each session provide a lasting learning experience. I find this work immensely rewarding!

Overall, my teaching approach enables me to seamlessly teach a variety of courses in operations management. I have developed a set of teaching skills that are transferable across student communities, be it the undergraduates, the MBA/MS students, Doctoral students or working executives. I have received many teaching awards including Best Course Taught (Supply Chain Management) and Favorite Professor by MSTM 2013 class at UIUC, and have been on the Excellent teachers list of UIUC for all seven years that I have taught there (2008-2015). I was also nominated for the Teaching Excellence Award for Undergraduate Teaching in 2012. My teaching ratings at Texas A&M continue to be high, with an average of 4.85 out of 5 - This includes undergraduate, MS, MBA and PhD courses.

SERVICE

I recognize that service is an integral part of an academic career. I have supported the broader research community through my efforts to support peer research review and academic conference organization. I served as the Senior Editor for Production and Operations Management journal (2018-2020) and an Associate Editor for the Journal of Operations Management (2016-2021). Additionally, I have been a reviewer for the following journals -- Management Science, Manufacturing & Service Operations Management and European Journal of Operational Research. I have been the session chair for the M&SOM cluster for four INFORMS conferences and three POMS conferences and have organized the Global Supply chain track for the POMS 2016 annual conference.

I served as the editor of POMS society magazine, *POMS Chronicle till 2020*. In this role, I tried to disseminate the learnings and happenings in the POM society. My favorite initiative has been one-on-one interaction with the POMS Fellows, wherein I travel to the Universities housing the POMS Fellows and interview them on their journey. This has been appreciated a lot within the POMS community.

An active role that I have played till 2023 is in the M&SOM society, where I was a member (and later chair) of the Diversity and Inclusion committee. As an initiative to make our society more inclusive and respond to issues surrounding sexual harassment and racism, I worked with the committee members to design a survey and elicit responses from the society members. We then had lengthy conversations in the M&SOM Annual meeting on how to deploy initiatives to address the challenges that society members indicated.

For my service, I received the Best Reviewer Award from POM Society in 2016, and Meritorious Service Award from the Manufacturing & Service Operations Management Society in 2018. I also received the Service award from POMS Society in 2023 for my editorial contributions.

At Texas A&M, I have also been actively involved in the Ph.D. program - in promotion of our program and in selection, recruitment and training of our Ph.D. students. I also continue to serve in a variety of roles that are directed at providing better teaching and learning environment for our students. In 2020, I worked with an internal team to redesign the

graduation experience for our undergraduates, onboarded the faculty for teaching on Zoom when COVID struck, and shared consulting tools, tips and tricks with our MBA students. I served on the campus-wide Associate Professor task force till 2022.

I am currently serving actively on three University Level Committees. First, I serve on the University Academic Freedom Council member (2024 - till date). We have had multiple meetings, where I worked with other members to deliberate on the current issues related to academic freedom at our University. Second, I serve on the University Honors & Undergraduate Research Advisory Committee (2024 - till date). In this committee also, we had multiple meetings, where I worked with other members to update the Honors requirements and relook at the undergrad research requirements. Third, I have been serving on the Faculty grievance committee since 2021. I have chaired the grievance committee to evaluate a faculty complaint: Multiple meetings with committee members and the faculty member were held. I asked for data from the related College, and examined documents, with follow-up meetings and advised the provost as to the complaint. This effort was noted as "Timely and Substantive" by Provost Office. I also worked on several cases with other members in 2024.

In outside consulting arena, I advised Acuva (acuva.com), an LED based water prurification firm. I worked with the CEO on issues of supply chain management, new product development and financial restructuring (2018-2024).

In my personal life, I am very actively involved with the education of underprivileged children, specifically with (a) Laxmi Ashram, an organization in Kausani hills in India, which works for the education of underprivileged girls and (b) Anjali School, located in the city of Varanasi in India, that works for the education of orphans and the poorest of the poor.

Goals (Faculty Evaluation Review)

Goals		
Please list what your goals were for the previous year and what your planned goals are for the upcoming year.	Start Semester	End Semester
My focus was on moving forward on the multiple research projects that are ongoing with collaborators. We advanced on several projects, and are nearing completion on a couple of them. I also completed my teaching duties and took on much more active service roles at the University level. I also chaired a session in the MSOM cluster at INFORMS 2024. For the coming year, my focus is on completing my teaching duties, serve at the University level roles, and focus on moving ahead with all research projects.	Spring 2024	Ongoing

External Consulting

External Consulting				
Client Name	Estimated Hours Involved	Description	Start Semester	End Semester
ACUVA	50	Advising the CEO on supply chain management, new product introduction, and financial restructuring.	Spring 2018	Spring 2024

Mandatory Training (Faculty Evaluation Review) - SECTION MUST BE COMPLETED

Certifications I certify my mandatory training is current and no outstanding training is pending:Yes I certify that I have no past due, outstanding or pending Concur items:Yes I certify that I have no past due, outstanding or pending Aggie Buy contracts: Yes I certify that I have no past due outstanding or pending Time and Effort items:Yes Start Semester: Spring 2024 End Semester: Fall 2024

Safety (Faculty Evaluation Review) - SECTION MUST BE COMPLETED

Safety in Teaching/Research Environment I certify that I am in compliance with all University/EHS safety requirements.:Yes Start Semester: Spring 2024 End Semester: Fall 2024